Formal Modeling of the Social Impact Theory: Applications to Social Loafing and Effort Matching phenomena

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Author Note

Author roles were classified using the Contributor Role Taxonomy (CRediT; https://credit.niso.org/) as follows: Samuel Riess: writing, conceptualization

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Abstract

In light of the replication crisis, precise links between theories and their emerging phenomena are becoming increasingly relevant for the psychological research community. With this research, we aim to test the connection between the existing Social Impact Theory and one of its emergent phenomena, the Social Loafing effect. Further, Effort Matching as an extending mechanism was observed to generate a broader perspective. Throughout our formalization, we used the Visual Argument Structure Tool to reshape the verbal tenets into a graphic illustration which contains all relevant identifiable constructs of the theory. The resulting mathematical model contained several extensions and restrictions, driven by the goal to design a self-consistent framework and by the need to address inconsistencies in previous research. Whilst simulating data for a fictional experimental design, we provided support for the model and therefore the Social Impact Theory itself. This leads to a logical conclusion that Social Loafing and Effort Matching emerge directly from its theoretical foundation, proving that this specific connection between theory and phenomenon requires no substantial modification.

Keywords: Theory Formalization, Social Impact Theory, Social Loafing, Effort Matching, Visual Argument Structure Tool (VAST)

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Objective

In this report, we formalize an existing psychological theory to test the relation between the emerging phenomena and the corresponding data by means of a computational model, based on the underlying theory. Following the methodological framework outlined by Borsboom et al. (2021), we generate simulated data for a fictional experiment inspired by the original study in which the phenomenon was first described. We are mainly interested in replicating key empirical findings associated with the phenomenon and to demonstrate that the model is capable of reproducing the phenomenon across different hypothetical scenarios. The need for such work is grounded in broader concerns within psychological science, known under terms like "reproducibility crisis" (Oberauer & Lewandowsky, 2019). Hence, our aim is to take a closer look at the relationship between the underlying theory and the emerging phenomenon, while evaluating how well the theoretical assumptions can account for the observed effects.

Introduction of the theory

We selected the Social Impact Theory because it provides a framework of clearly operationalized variables, allowing a precise measurement for input factors and their resultant phenomena. While the theory was first introduced in 1973, Latané published the full framework of mechanism and effects eight years later (Latané, 1981). In the meantime, the foundational definition of 'social loafing' was introduced (Latané et al., 1979), using the tenets of the theory to explain the phenomenon. I added more depth in referring to an extending explanation of the mechanism, named 'Effort Matching' (Jackson & Harkins, 1985).

Core constructs

Our formalization is grounded on core constructs derived directly from primary research literature. Some constructs were specified inconsistently over the research, so we aimed to label them in a way that serves a greater interpretation. Of course, these labels are our own understanding of the constructs, other researchers may focus on other elements. The scope is highly limited to the main social loafing research, we only differ if a special need or

cause occurred. The central social impact theory extends beyond our chosen phenomenon which falls outside our scope. All constructs and their sources can be seen in (Table 1), our presented VAST model in Figure 2 contains them in a connected and graphic way to understand the underlying mechanism of the phenomenon. As one can see, the initial constructs that cause the effect are the Co-Targets (CT) and Sources (S) of social impact, presenting group members (e.g. experimenters who supervise the group in an experiment). The sources can be every kind of person that performs pressure or impact on a certain group or individual. On this note, it is worth mentioning that pressure and impact are not further separated so we're sticking with the definition "pressure" in this report, resulting in our construct P. The contingency between input and outcome (CON) marks a prior expectation, that's why its presented as part of the "Psy"-Box. It leads later to the "incentive to work equally hard" (INC), linked through a proportional relationship (Latané et al., 1979). Their values are the same, whilst INC contains a more motivational nature. The final psychological output of the mechanism is the individual effort (IE) which is the cause for the manifest output later on. The constructs "effort matching condition" (EMC) and "loafing expectation" consist both of three conditions, marking the 3 conditions of Effort matching I implemented in the research from another source (Jackson & Harkins, 1985). We didn't include constructs like "identifiability"" due to our strict replication of Latané et al. (1979) and setting identifiability to 'not given', based on the original experiments (Latané et al., 1979). The same reason accounts for disregarding the constructs immediacy and strength of S or CT.

Identification of the relevant Phenomena

In Latané et al. (1979), the resulting social loafing is described as follows: 'People exhibit a sizeable decrease in individual effort when performing in groups as compared to when they perform alone'. The extending Effort Matching this phenomenon is defined in another research, stating: 'they will maintain equity by matching the effort levels they expect from their partners' (Jackson & Harkins, 1985). This was already discussed before (Latané et al., 1979) but further investigated in the newer research. The idea is that social loafing can be eliminated if the expectation to loaf, drawn to the effort of other group members, is held constant and therefore blocks the initial psychological process.

To create a link between this phenomenon and our formalization of the Social Impact Theory, we reapplied our visual framework. The manifestation of this phenomenon is illustrated in Figure 1.

To address the robustness of social loafing and justify the need for further investigating, our analysis is divided into two distinct sections: Strength of evidence and generalizability, using the UTOS framework. We get our overview for both by analyzing a Meta-Analysis concerning social loafing (Karau & Williams, 1993) which conducts about 78 studies of the topic. We found quotes like 'using a range of subject populations varying in age, gender, and culture'. This leads to a variation of Units (u). The variation of treatment (T) is supported by using different attributes of the task itself, e.g. complexity, type of effort, quantity or quality. Within- and between-subjects comparisons across coactive and collective conditions took place in investigating social loafing. The variation of Outcomes (O) is supported by operationalizing the dependent variable through cognitive of physical effort. sometimes, it was based on an individual or group level. Finally, the setting (S) of the experiments took place in different countries, which we interpreted as a generalization over different cultures and a confirmation for the last UTOS-dimension. The strength of evidence can be acknowledged with mean effect sizes that 'differed significantly from the 0.00 value that indicates exactly no difference'. The most pithy evidence can be found in the result section. Directly, the mean effect size was found to be $d = .44$ which was statistically significant ($p < .001$), presenting a moderate effect according to Cohen's general conventions. Taken together, the phenomenon can be considered robust because the whole UTOS framework and the strength of evidence are supported with empirical data.

Method

All documents used in the process of writing this report can be found and downloaded from the GitHub repository: https://github.com/SamRiess/SocialLoafing_FINAL.git. The working Google Doc which contains all tables is linked here: https://docs.google.com/document/d/1uyZClUMjVYVrUacyiwE5CBqVdFDZlJFDtxe1ZKb-B74/edit?usp=sharing.

used software

Throughout our formalization, we used the Visual Argument Structure Tool (VAST, (Leising et al., 2023)) to build a graphic visualization of the theory and resulting mechanisms. With this methodology, we were able to define relationships, our understanding of core constructs and the mechanisms that take place in the process of social loafing and effort matching. For the computation of our simulated data, we used the statistic software R. The VAST model was constructed withing a draw.io environment.

Formalization of the (Proto-)Theory

In the process, it was crucial to limit our scope, referring to both theory and phenomenon. The main goal is understandability, not the whole integrity of the full theory. Our scope was limited to the following aspects: As noted earlier, we set the identifiability of the individual effort to non-existent. This can be justified because identifiability was not part of the experiments for social loafing (Latané et al., 1979), neither was it for effort matching (Jackson & Harkins, 1985). Further, we ignore the multiplicative connection between immediacy, strength and number of CT or S (Latané, 1981). These aren't, again, part of previous research so we did not focus on that specific part of social impact theory. A few more additional assumptions were crucial for our finalized theory and model to work. Other mentioned constructs like the default expectation for other peoples loafing wasn't specifically evaluated and included in both researches. We addressed this in using the mechanism in the Psy box to generate a default expectation. With this approach, the underlying process gains more understandability while the high and low Effort condition (HE and LE) could be included into the superfunction for our psyche process without complicating the model itself. The extension of the maximum capacity was necessary for simulation purposes, it isn't mentioned in the theory itself either. the inclusion of MC was essential to operationalise a manifest variable IO as the primary output of the underlying psychological mechanism results in percentage. This value requires a baseline representing 100% of potential output to remain measureable in our simulation. See Table 2 for further values. As we're introducing later in this report, these restrictions or extensions weren't detrimental for our Model and replication of the loafing effect and are therefore providing a post-hoc evaluation. The resulting prototype

theory which ought to explain the phenomenon is presented in Figure 2. The 'relationships' section is showing each relationship and its justification, based on quotes from previous research (labeled with a "q" in Table 3) or our own assumptions and restrictions. In keeping this section short, I am mixing up the explanation of the model with its justification of the described relationships.

Variables

All used variables for each construct, if shown in our VAST, can be reviewed in Table 2.

Relationships

All relationships are mentioned in Table 3 with the relevant quotes, labeled with a "q" in the following section. The relationship between CT, S and P is presented in Figure 3. We aimed to write a function that is consistent with exact quotes from the sources. Originally, the pressure can be lowered throughout the number of CT but increased with more S. As we can see in the graphs, the higher the number of S is, the higher gets the felt impact on one subject. This represents the classic increase which is mentioned in q4. But with an increase of the number of CT, the impact lowered rapidly. Therefore, consistent with the general interpretation of social impact theory, felt impact is divided up among the group members (q5-q8). No specific expectation about the weight of each process is given, so we filled this gap with our own values.

The left path of our VAST model (Figure 2) illustrates the primary mechanism underlying the phenomenon. It starts with the relationship between CT and CON which was rather intuitive due to the specific function derived from the core report, verified with (Latané et al., 1979) and specifically in q1-q3. It is represented in Figure 4.

After that, the transformation into INC can be transposed by a proportional increase: INC gets the same values as the previously computed CON (Figure 5). This was not fully based on a focal theory, but we infered that an incentive to give one's own full effort directly results from the depicted CON, also supported with q14.

Things were getting more interesting with the function that computes the resulting IE, using all previously simulated and processed input variables. Both paths of the VAST meet together in this relationship. The plot can be viewed in Figure 6, showing the first influence of

EMC as well. I wrote a function, which transforms the manifest EMC into the latent construct LE, named t3 and supported by q12/q13. If the subject takes action in the LE (low effort) or HE (high effort) condition, effort matching takes place and fixes individual effort to a previously set value for each condition. While these values can be adjusted independently, they were fixed to ensure the HE condition exceeds the LE condition under all circumstances. No further information about these values was found in the research, so this addition was necessary (Jackson & Harkins, 1985). The 'no_info' graph shows the natural social loafing mechanism for the individual effort, based on no information whatsoever about the others effort. I used it for the replication condition SLR. If the pressure is high, the function starts higher cause the subject feels more bound to the outcome. Secondly, with more INC the subject is getting higher values in the effort. We argue that, besides great pressure, the effort won't reach its maximum if INC isn't maximized. The basic idea of the relationship is supported by q9-q11. to align with my theoretical framework, I adjusted the function for IE, assuring P and IE range between 0 and 1.

The critical relationship for the simulation is the transformation of IE and MC into IO. As can be seen in Figure 7, the more MC each subject possesses, the higher the values for IO get. That occurs cause we defined IE as a percentage of 1 (e.g. 0.6 = 60%) which is offset against MC. Hence, with more IE, IO is increasing simultaneously. We concluded an interaction effect between MC and IE that accounts for the final value of one's own output. The transformation function was needed to compute the manifest outcome for each person (e.g. in q15) and the group means for our following statistical analysis.

The behavior of the combined system is shown in Figure 8. It's similar to the IE function but deviates with its own input variables. In this case, we used all manifest ones and incorporated them into the whole psychological process that's labeled "Psyche" (see Figure 1 for clarification). The resulting "superfunction" can be described as followed: If the effort matching condition it set to low effort (1) or high effort (3), the function fixes IE to a given value. This is independent from the number of S, whilst in the replication condition (2) the classic social loafing effect takes place, remaining highly contingent upon the number or S.

Evaluation of the formal model

We simulated the original experiment for social loafing, using the sound pressure as the individual output (Latané et al., 1979) but added the EMC predictor, presenting the "Effort matching condition". With a sample size of n=36 (view Latané et al. (1979), we were able to test for the original effect and the effort matching mechanism in one sample). The descriptives of both analyses are showcased in Figure 9 and Figure 10. These are highly influenced by the noise term of p3 which can be adjusted freely, for the original research didn't provide such values. We chose the parameters mean = 0 and sd = 0.2 for our 'noise term' of that function, cause we assume that variability comes into the model throughout many sources: e.g. fatigue, personality or the current mental state.

Due to 2 computed ANOVAS which rely on the same sample, we needed to adjust the p-value with the Bonferroni method. Thus, the new α-level comes down to .025. It turns out that both effects, on one side the social loafing but on the other the elimination of this phenomenon, based on effort matching, were supported by our present simulation and therefore our model. In proving the results, we fall back on the original analysis (Latané et al., 1979) for which all resulting graphs can be viewed in Figure 11. The main replication was significant for the number of Co-Targets had a significant effect on the individual sound output $F(2, 105) = 51.24$, $MSE = 6.51$, $p < .001$, $\hat{\eta}^2_G = .494$. This is consistent with the findings of Latané et al. (1979) and proves that the model is capable of producing the main phenomenon. Whilst replicating the findings of effort matching (Jackson & Harkins, 1985), I compared all three pairs of 'Alone vs. Group', each for one condition of EMC. Only in the SLR-condition, I detected a significant decrease in the outcome between the group and alone situation (p < .001). For the low effort condition the difference is no longer significant with p=.323, equal results appeared in the the high effort condition with p=.223. Therefore, I've shown that Effort Matching emerged via our model and is consistent with the literature in eliminating the social loafing effect thoroughly holding the loafing expectation constant.

Meta-reflection

Some issues need to be discussed at the end. First, the original paper contained many jangle-fallacies, so we had to form clear definitions for our main constructs. An alternative

VAST image (Figure 12) illustrates this conceptual separation, highlighting that each construct should be clearly defined without redundant or overlapping terminology. Second, it was a challenge to implement the sources (S) into our model, as these factors were conceptualized in the underlying theory but not explicitly manipulated within the experimental design (Latané et al., 1979). A clear limitation of the scope was essential for formalizing this theory while using VAST. Hence, the mentioned inconsistencies in the original experiment and its explanation had to be managed. In regard of the simulation, the decision if the VAST itself was to contain a group level was partially difficult - in the end we decided to stick with the individual output as the final construct. To establish a group level directly into the VAST, we would have needed to adjust the simulation by redefining the CT construct as 'group members. This would have necessitated a whole new structure of the entire psychological mechanism of social loafing.

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Appendix

Concept Table - Social Loafing
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Table 1

Table of core constructs

ID	Type - Phenomenon - Concept - HOC	Short name	Quote	Reference	Rel. type (n, p, i, r, ...)	Comment (e.g., identified gaps & inconsistencies)	Incl. (Y/N)?
CT	C	Co-Targets	-"The more people who are the target of this pressure, [...]" (S.823) - "According to Latane's (1981) social impact theory, people can be viewed as […] targets of social impact." (S. 682) AND - "and number of group members" (Latané et al., 1979, S. 823) - "impact is divided among the group members" (Latané et al., 1979, S. 830) - "However, if a person is a member of a group that is the target of social forces from outside the group" (Latané et al., 1979, S. 823)	Latané et al., S.823 (1979) AND Karau & Williams, S. 682 (1993)	n1, p1,p2, n12		Y
S	C	Source of pressure	- "The impact that the experimenters have on an individual [...]" (S.825) - "According to Latane's (1981)	Latané et al., S.825 (1979) AND	n2, p2	these persons are not described as a pressure precisely,	Y

			social impact theory, people can be viewed as either sources […] of social impact.” (S. 682)	Karau & Williams , S. 682 (1993)		but we can assume that they are the reason why the pressure occurs we set: pressure = impact	
CON	C	Contingency between input and outcome	“[...] the contingency between their input and the outcome [...]” (S. 830)	Latané et al., S. 830 (1979)	p1, n3, t1	here, contingency is simply explained through the word itself, no further definition	Y
INC	C	Incentive	“[...] incentive to work equally hard in groups of different sizes.” (S.830)	Latané et al., S. 830 (1979)	t1, n4, p3		Y
P	C	impact/ pressure	“the amount of social pressure on the target person” (S. 823) “pressures on each individual” (S. 830) / “In a group performance situation in which pressures to work come from outside the group” (S. 830) AND “[...]the impact of these forces on any given member [...]” (S.823)	Latané et al., S.823 (1979)	p2, n5, n6, p3	impact and pressure are no really distinctive concepts, cause impact can officiate as a social facilitation. here, we assume its the same cause we limited the	Y

						scope to the negative form of social impact, no facilitatio n at all	
IE	C	individual effort	“individual effort” (S. 822 ff.)	Latané et al., S.822 ff. (1979)	p3, n7, t2	In this paper, there is no real distinction between “effort” and “performa nce”. These concepts were used partially synonymo usly. The problem is, they were not defined.	Y
Psy	HO C	Psyche	no citation	/	n11	We defined this HOC as the space in which the mental process is defined	Y
IO	C	individual output	“in this case n times the average individual output” (S. 825) AND “Since individual scores” (S. 830) AND - “leading to an apparent drop in	Latané et al., S.825/S. 829/S.83 0 (1979)	n13, n14, n15, t2		

			individual performance, a phenomenon we call social loafing" (S. 825) - "sum of their individual performances" (S. 829)				
MC	C	maximum capacity	"dyads pulled at 91% of the sum of their individual **capacities**, trios at 82%, and groups of six at only 78%" (S. 825)	Latané et al., S.825 (1979)	n16, t2	maximal capacity of individual, when 100 sources and 0 cotargets are present (no citations are available, this is our own interpretation to simulate the phenomenon)	Y
ME	HOC	Matching of effort	"this effort matching interpretation" (S.1204) "they will maintain equity by matching the effort levels they expect from their partners" (S.1200) "social loafing occurs because individ uals expect others to slack off in groups and, therefore, reduce their own efforts to maintain equity" (S.683)	Harkins & Jackson, S.1200/1204 (1985) AND Karau & Williams, S. 683 (1993)	/	this effect occurs only when the scope is limited to holding identifiability constant throughout the experiment	N
EMC	C	Effort condition	"In the HE condition, she said she had tried very hard on the practice trial and, because she thought the research was interesting, she was	Harkins & Jackson, S.1202	n17, t3	this is purely a condition construct,	Y

			going to try hard throughout the experiment; in the LE condition, she said she had not tried on the practice shout and, because she thought the research was boring, she wasn't going to try hard during the rest of the experiment; and in the SLR condition, she simply made small talk until the experimenter returned" (S.1202)	(1985)		no explicit values were give in Harkins & Jackson (1985), manifest variable outside the Psy-Box	
LE	C	loafing expectation	"expectations of the amount of partner effort" (S.1204) AND "expectations concerning the partner's intended levels of effort"	Harkings & Jackson, S.1204 (1985)	n18, t3, p3	no values were given here either, but this is strictly a latent variable - we transform it from the effort matching condition to have a construct that matches the condition in the Psy-Box	Y
SL	P	Social loafing	"people exhibit a sizeable decrease in individual effort when performing in groups as compared to when they perform alone"	Latané et al., S.822 (1979)	i1	this is the core phenomenon for the report	

Table 2

Table of variables.

Construct	Source (if given)	Scale level	Range/Values	Anchors
CT	"and *number* of group members" (Latané et al., 1979, S. 823)	discrete	$\in \mathbb{N}_0$	0 = "zero cotargets are present"; n = "number of cotargets (other persons in the evaluated group, number of group members minus 1)"
S	"*number* of these other persons." (Latané et al., 1979, S. 823) AND "Thus, as the *number of performers decreased, the size of the audience increased*. According to Cottrell's evaluation apprehension hypothesis (1972), the presence of an evaluative audience should enhance performance for a simple, well learned task such as rope pulling, and, although there is little supportive evidence, it	discrete	$\in \mathbb{N}_0$	"0 = "zero persons that cause the social pressure"; n = "number of persons that cause the pressure"

	seems reasonable that the *larger the audience*, the greater the enhancement“ (Latané et al., 1979, S. 826)			
P		continuous	∈ [0; 1[	0 = “zero persons are source of social pressure”, “no social pressure” 1 = “maximal social pressure (a theoretical amount on infinite spectators are present)”
CON	“Individuals could "hide in the crowd" (Davis, 1969) and avoid the negative consequences of slacking off, or they may have felt "lost in the crowd" and unable to obtain their fair share of the positive consequences for working hard.” (Latané et. al, 1979, S. 830) “reward” (Latané et. al, 1979, S. 830)	continuous	∈ [0; 1]	0 = “expectation that the share (reward/punishment) of each member is completely independent of what they’ve given” 1 = “expectation that their share (reward/punishment) is equal to what they’ve given”
INC		continuous	∈ [0; 1]	0 = “no incentive to give any individual effort” ; 1 = “incentive to give full effort”

IE	e. g. “dyads pulled at 91% of the sum of their individual capacities, trios at 82%, and groups of six at only 78%” (Latané, 1979, S. 825)	continuous	∈ [0; 1]	0 = “the person shows no effort at all to bring in some contribution to the group task”, 1 = “the person shows their maximal effort in contributing to the group task”
O	“sum of their individual capacity” (Latané, 1979, S. 824); e. g. “dyads pulled at 91% of the sum of their individual capacities, trios at 82%, and groups of six at only 78%” (Latané, 1979, S. 825)	continuous	∈ [0; 1]	0 = “maximal social loafing effect, nobody does anything”; 1 = “sum of the maximal individual performances” “no social loafing effect” “everybody gives their best” o = “percentage of the theoretical maximal outcome (sum of each individual capacity) of the whole group”
IO	“dyads pulled at 91% of the sum of their individual capacities, trios at 82%, and groups of six at only 78%” (Latané, 1979, S. 825)	continuous	∈ [0; inf[	0 = dynes/cm^2 n = dynes/cm^2
MC		continuous	∈ $\mathbb{R}_0$	depends on task (clapping with hands, rope pulling)
EMC	“she said she had tried very hard	ordinal	1, 2 or 3 is possible	1 = low effort condition,

	on the practice trial and, because she thought the research was interesting, she was going to try hard throughout the experiment" (Jackson & Harkins, 1985, S.1202) AND "she said she had not tried on the practice shout and, because she thought the research was boring, she wasn't going to try hard during the rest of the experiment" (Jackson & Harkins, 1985, S.1202) AND "she simply made small talk until the experimenter returned" (Jackson & Harkins, 1985, S.1202)			2 = Social loafing replication condition, 3 = high effort condition
LE	view "EMC" for citations, they're identical	ordinal	"match_low", "no_info", "match_high"	"match_low" = the subject thinks the Co-Targets were giving their full effort in the practice trial, "no_info" = the subjects has no information whatsoever about the others goal in effort,

“match_high” = the subject thinks the Co-Targets had not tried hard in the practice trial

Table 3

Table of relationships.

Output variable	Function to compute the output variable	Comment	To what extent derived from theory?[a]
relationship P1 => CON	CON = 1/(CT+1) => CON = 1/n	quote q3 mentions the "reward" but if each group member performed up to its own capacity would feed an inconsistency between the effort (100%) and the reward/outcome (1/n)	CT → CON: q1: "We believe that social loafing occurs because there is usually a stronger perceived contingency between individual effort and valued outcomes when working individually. When working collectively, factors other than the individual's effort frequently determine performance, and valued outcomes are often **divided among all of the group members**." Karau & Williams, S. 684 (1993) q2: "which both suggest that individuals' outcomes are more contingent on their efforts when working with others who are expected to perform poorly." Karau & Williams, S. 696 (1993) q3: "the reward is psychologically divided equally among the participants, each getting 1/n units of reward" (Latané, 1979, S.830)
relationship P2 => P	P = S/(S+CT+1)		q4:"[...]the amount of social pressure on the target person should increase as a multiplicative function of the strength, immediacy, and number of these other persons." (Latané, 1979, S.823)

			=> derived from theory q5: “Impact is divided up among the group members, in much the same way that responsibility for helping seems to be divided among witnesses to an emergency” (Latané, 1979, S.823) => derived from theory q6: “If a person is the target of social forces, increasing the number of other persons also in the target group should diminish the pressures on each individual because the impact is divided among the group members.” (Latané, 1979, S. 830) CT → P: q7: “The more people who are the target of this pressure, the less pressure should be felt by any one person.” (Latané, 1979, S. 823) => derived from theory q8: “increased group size to result in reduced effort” (Latané, 1979, S. 823)

relationship P3 => IE	get_IE <- function(INC, P, LE) {
	  # the Social Loafing baseline (Latané)
	  # whilst CT increases, the output decreases
	  loafing_effort <- 0.5 * INC + 0.5 * P
	
	  # the effort matching logic
	  # we're definign fix values which represent the expectation for the "alone"-
	  # condition of other
		we added the effort matching condition/variable later on, no real mathematical relation could be found in the underlying paper
		
		this function includes one of two sources of variability between the subjects cause
			q9: “Latane further suggests that just as psychophysical reactions to external stimuli can be described in terms of a power law (Stevens, 1957), so also should reactions to social stimuli, but with an exponent having an absolute value less than 1, so that the nth person should have less effect than the (n - l)th.” (Latané, 1979, S. 823)

subjects (independent of the current group size)

```
  final_IE <- case_when(
    # condition: co-Target is not loafing in its "alone"-condition
    LE == "match_high" ~ 0.9,

    # condition: Co-Target is loafing in its "alone"-condition
    LE == "match_low"  ~ 0.4,

    # In der no_info Bedingung greift das klassische Modell (Loafing)
    LE == "no_info"    ~ loafing_effort,

    TRUE               ~ loafing_effort
  )

  return(final_IE)
}
```

we added the noise term on top of it (but only in the superfunction version of it)

For this function, it was reasonable to present the full R code.

=> derived from theory

P → IE:
q10: “The more people who are the target of this pressure, the less pressure should be felt by any one person. Since people are likely to work hard in proportion to the pressure they feel to do so, we should expect increased group size to result in reduced efforts on the part of individual group members.” (Latané, 1979, S. 823)

q11: “If a person is the target of social forces, increasing the number of other persons also in the target group should diminish the pressures on each individual because the impact is divided among the group members. In a group performance situation in which pressures to work come from outside the group and individual outputs are not identifiable, this division of impact should lead each individual to work less hard. Thus, whether the subject is dividing up the amount of work he thinks should be performed or whether he is dividing up the amount of reward he expects to earn with his work, he should work less hard in groups.” (Latané, 1979, S. 830)

q12: “in fact, participants matched their own efforts to these new expectations” (Jackson & Harkins, 1985,

			S.1199)
			q13: “by matching the effort levels they expect from their partners” (Jackson & Harkins, 1985, S.1200) => derived from theory
relationship t1 => INC	INC = CON		q14: “Thus, under these assumptions, if his own performance cannot be individually monitored, an individual's incentive to perform should be proportional to l/n.” (Latané, 1979, S. 830) => derived from theory
relationship t2 => IO	IO = MC * IE	since we need so compute a manifest outcome for each person and therefore for the experimental group means we had to multiply the percentage of individual effort with the maximum capacity	q15: “as with rope pulling, output can be measured in simple physical units that make up a ration scale” => derived from theory

relationship t3
=> LE
	get_LE <- function(EMC) {
	LE <- case_match(EMC,
	1 ~ "match_low",
	2 ~ "no_info",
	3 ~ "match_high"
	)
	return(LE)
	}
		we transformed the manifest construct/variable EMC into the latent construct/variable LE what created the need for such transformation	=> Not based on focal theory, but rather on common sense or other theories

[a] *Response scale:*

- *Dictated by theory (add IDs from the Construct Source Table as reference)*
- *Derived from theory*
- *Loosely inspired by theory*
- *Not based on focal theory, but rather on common sense or other theories*
- *Not at all*

Figure 1

Structure of the VAST phenomenon

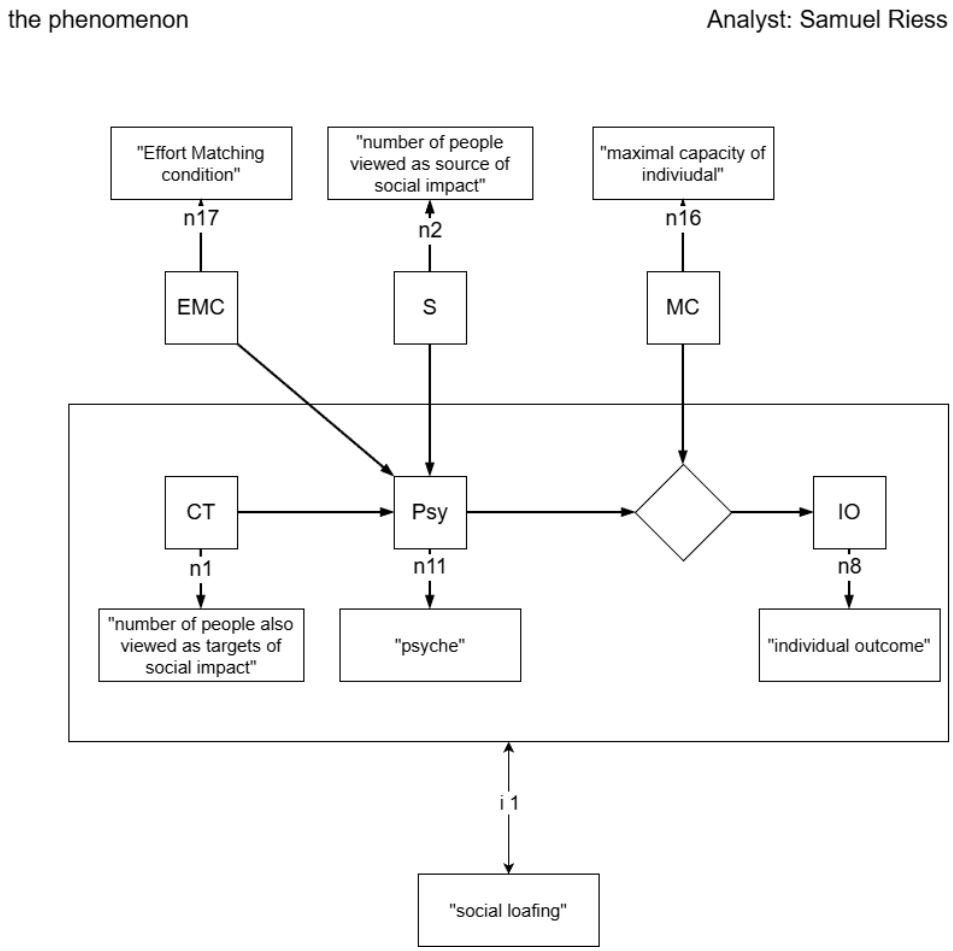


Figure 2

Structure of the VAST theory and mechanism

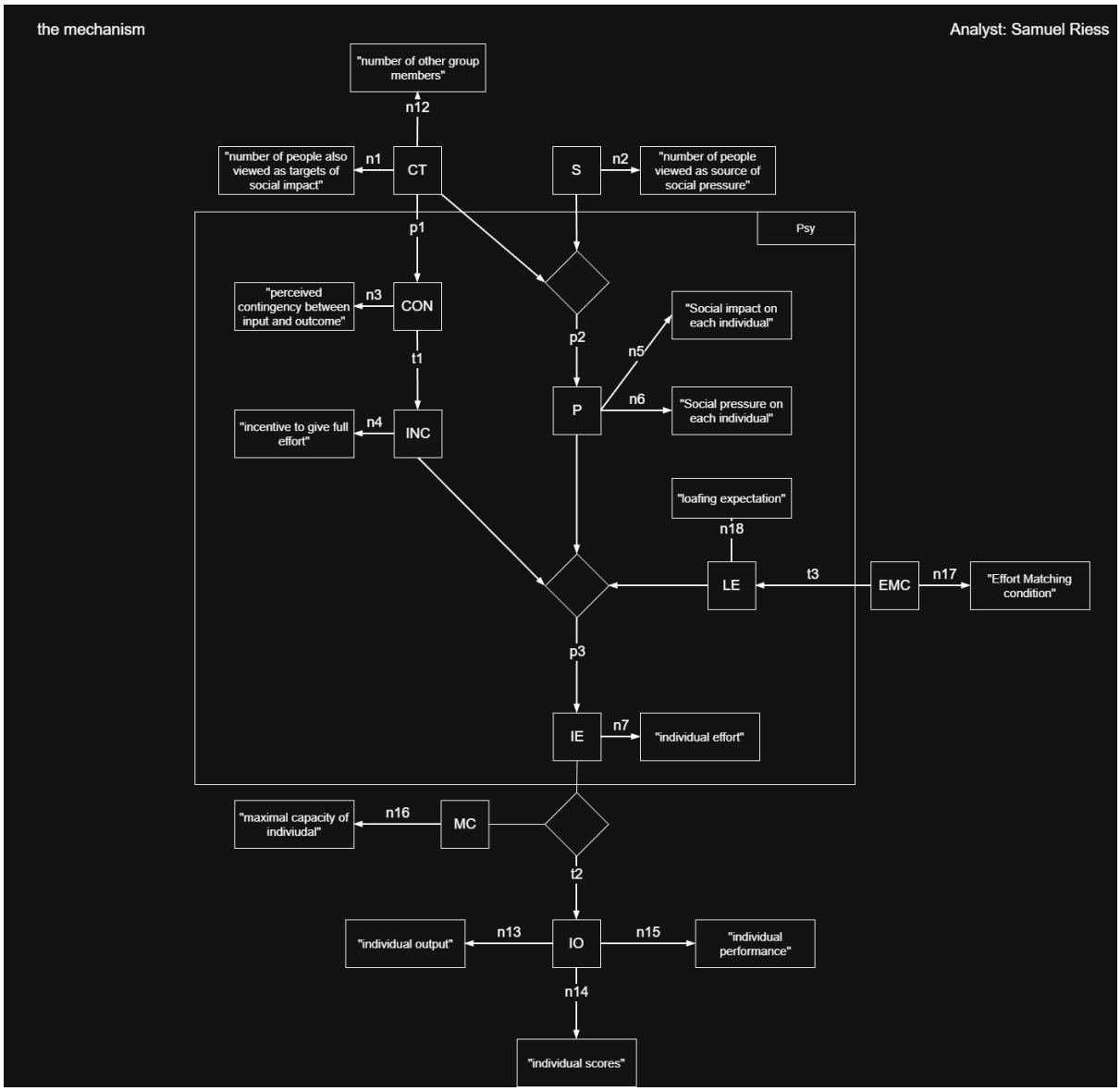


Figure 3

Plot for the relationship p2

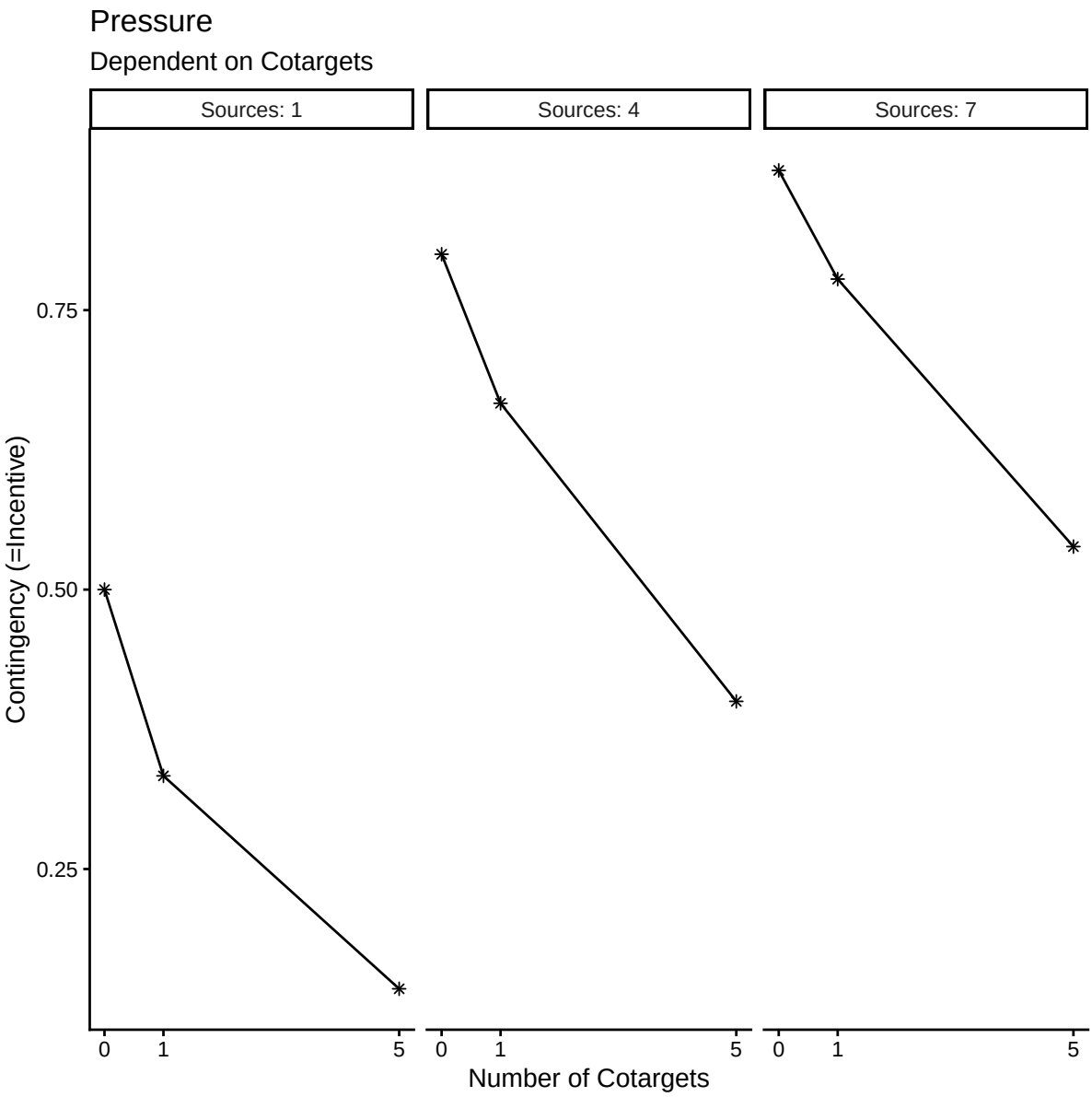


Figure 4

Plot for the relationship p1

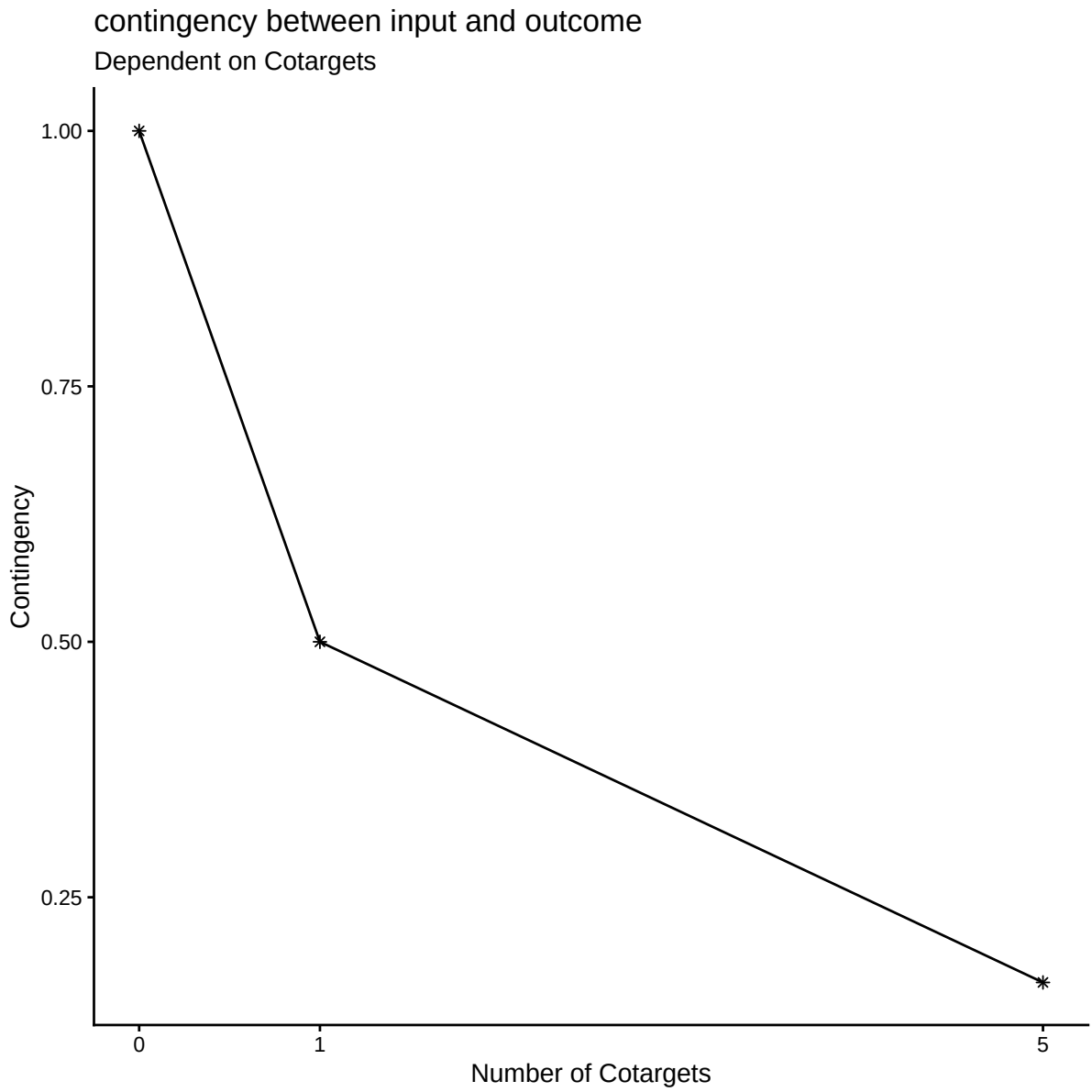


Figure 5

Plot for the transformation t1

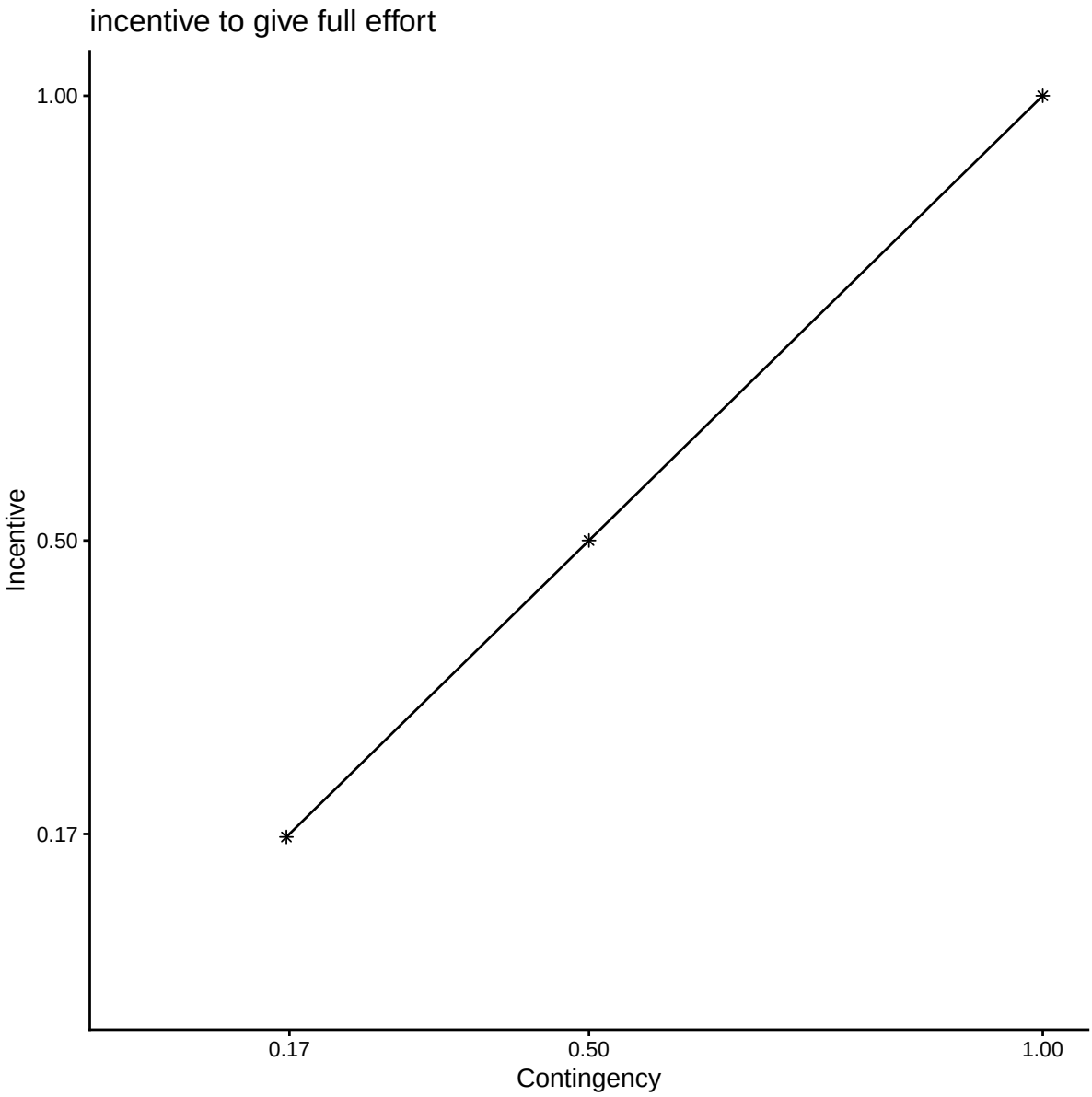


Figure 6

Plot for the relationship p3

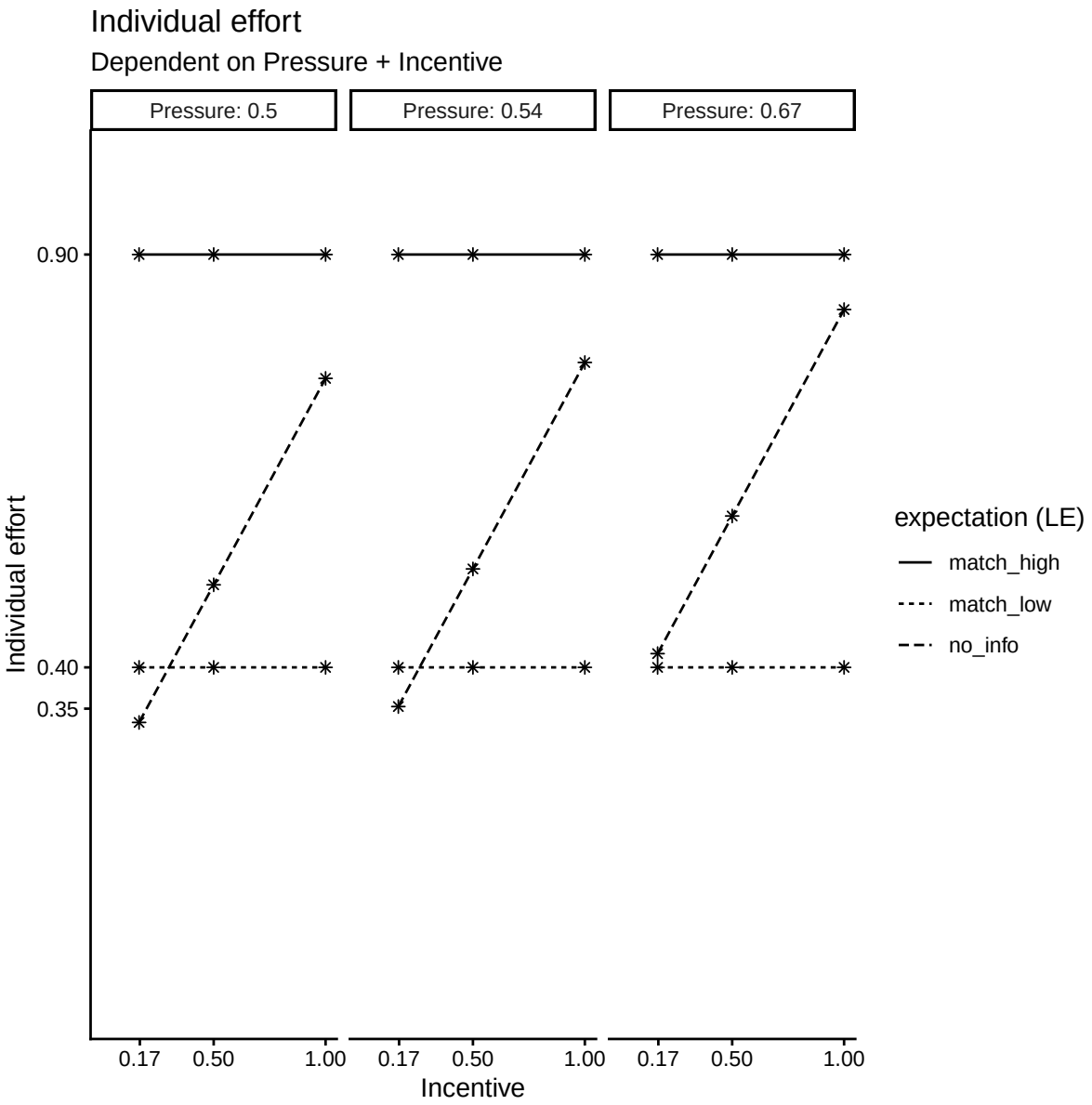


Figure 7

Plot for the relationship t2

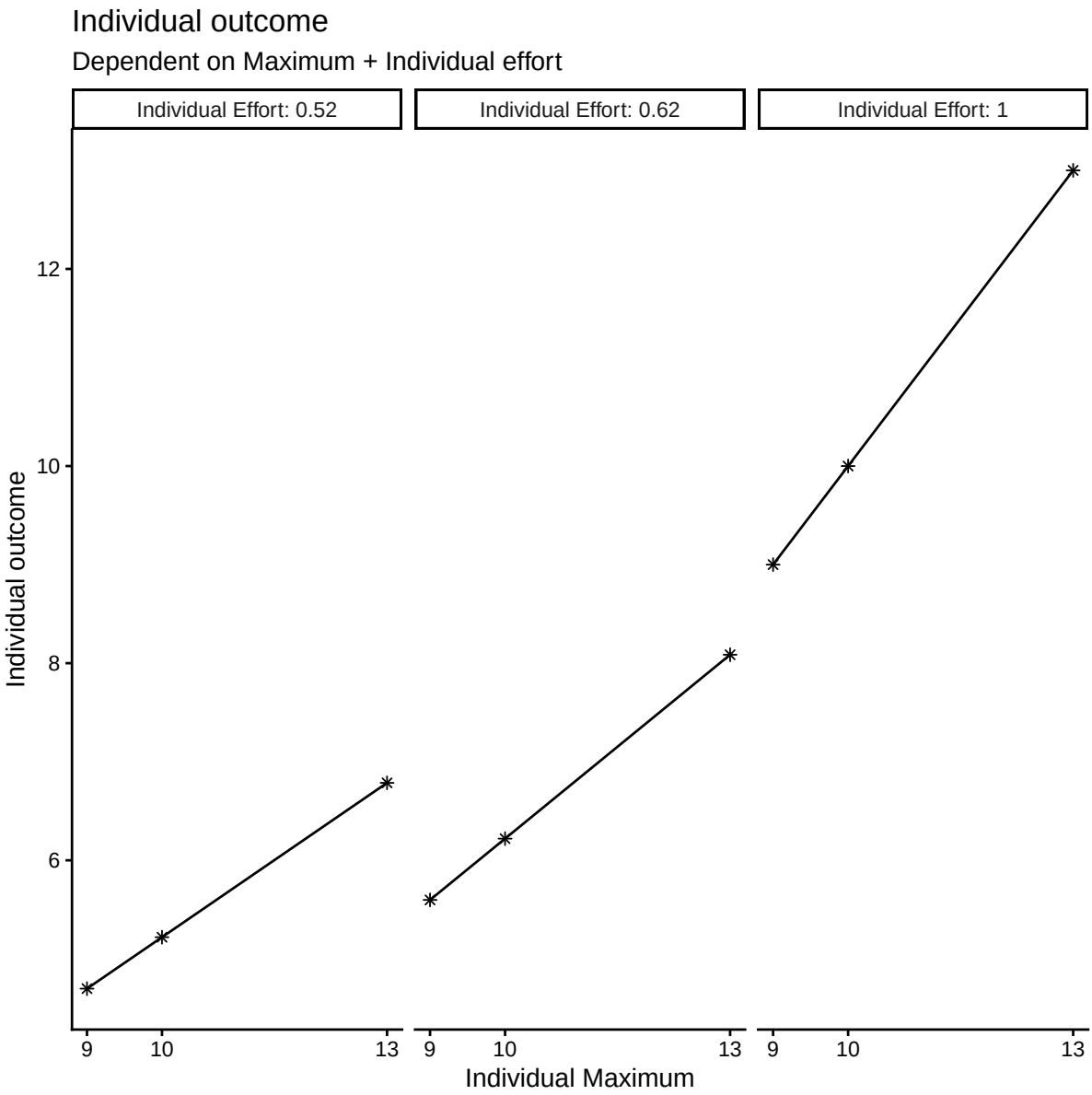


Figure 8

Plot for the resulting superfunction

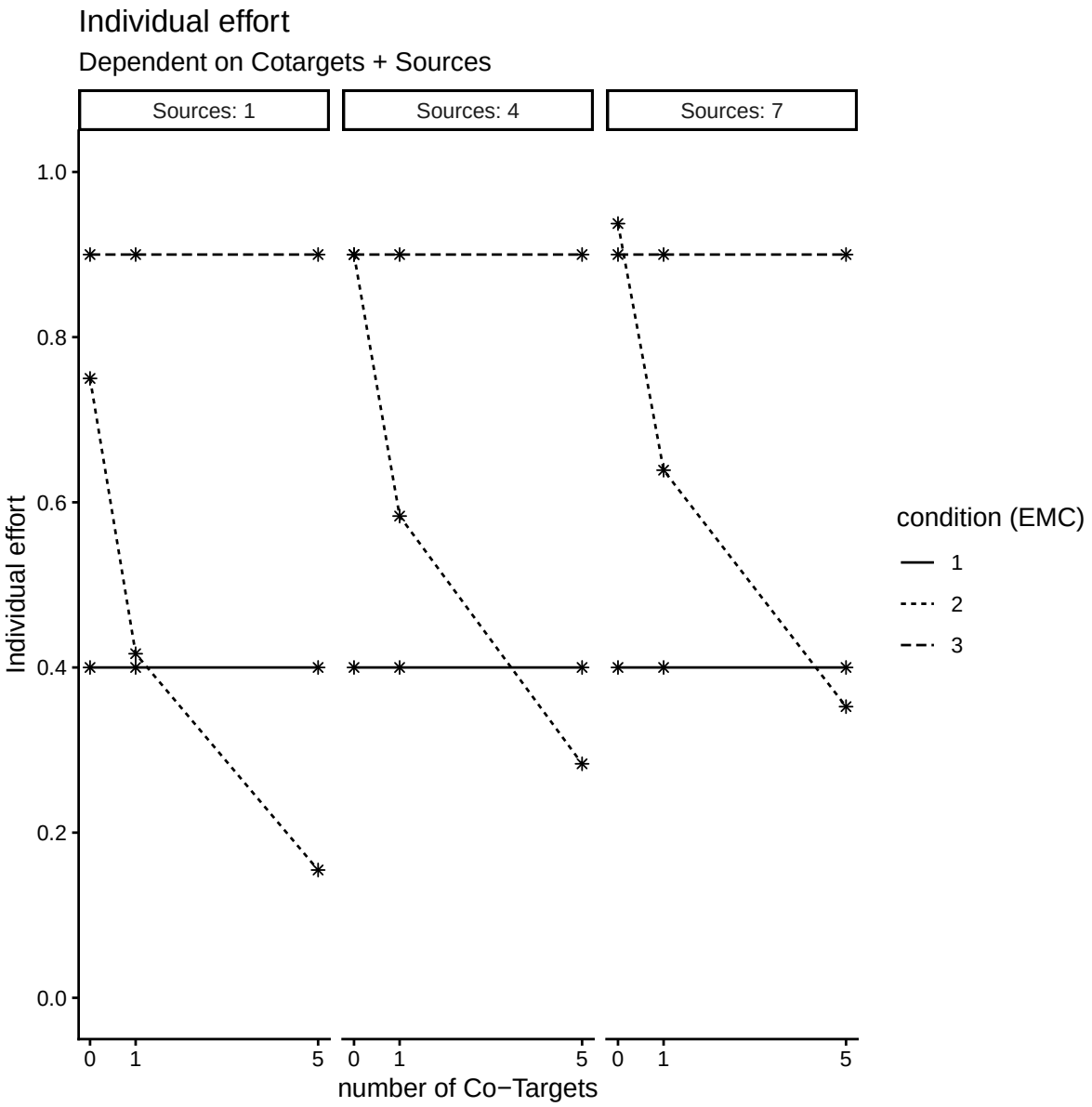


Figure 9

Boxplot for the replication of Latané

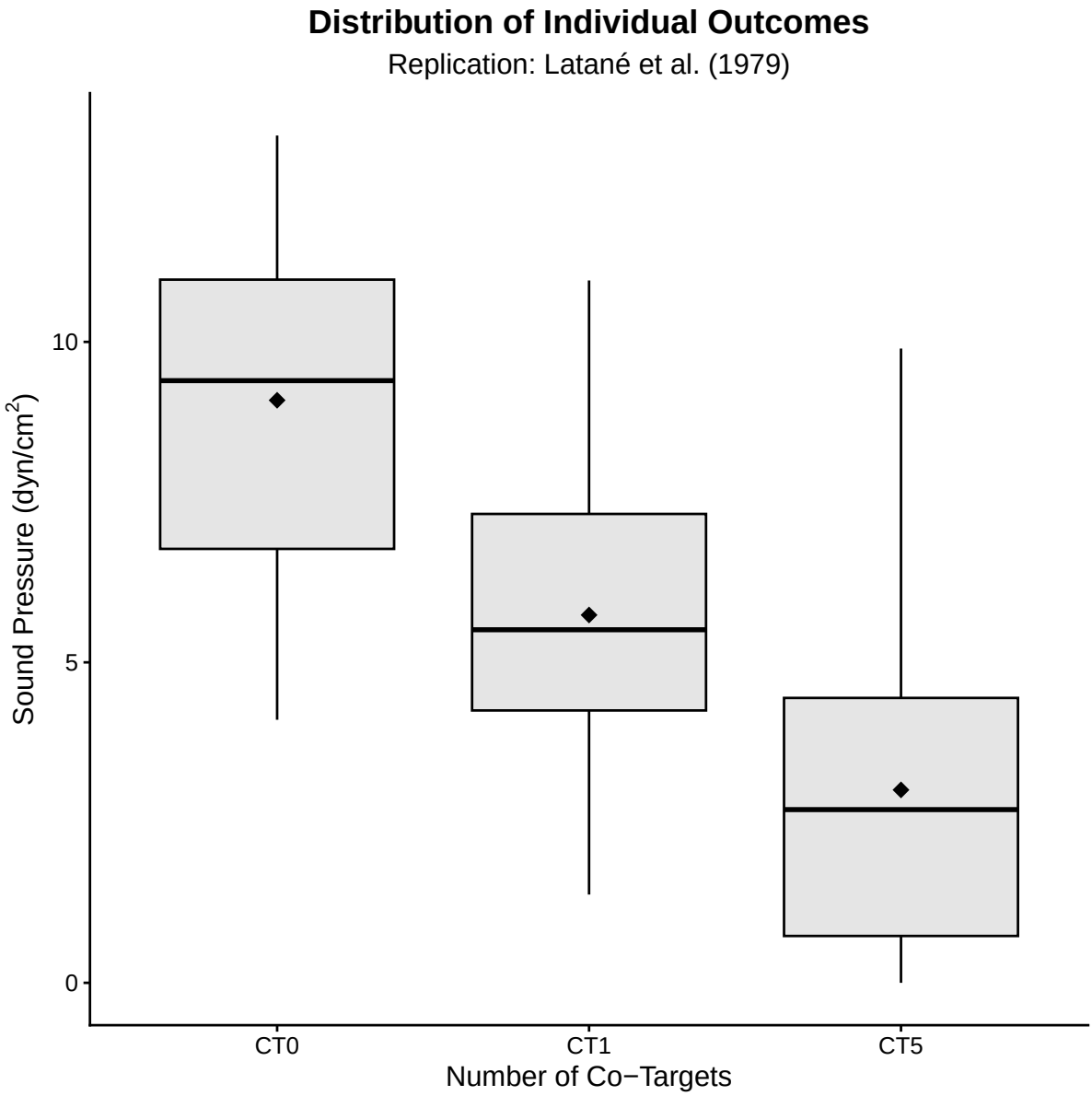


Figure 10

Boxplot for the replication of Jackson & Harkins

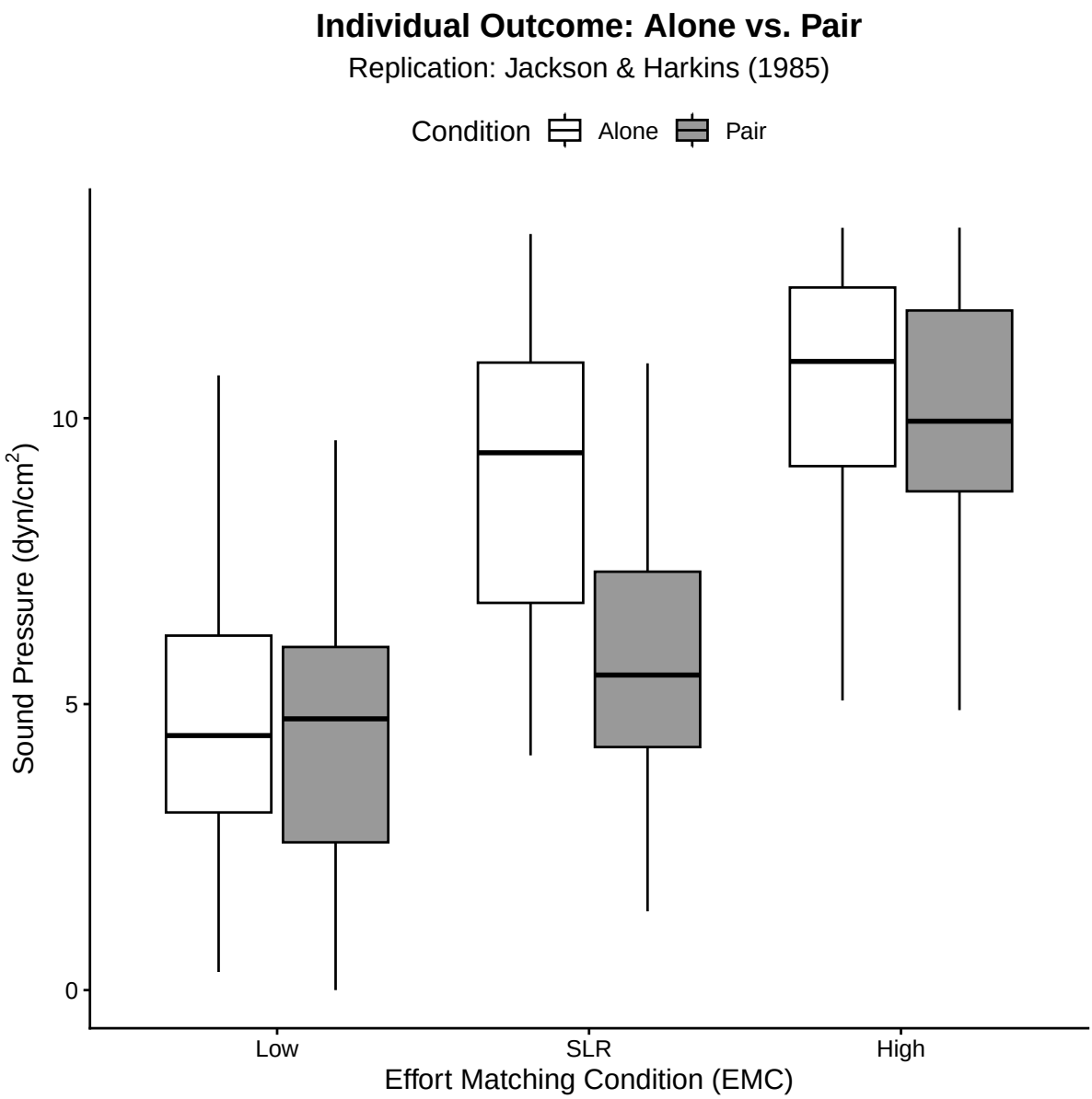


Figure 11

Final plot of the simulation

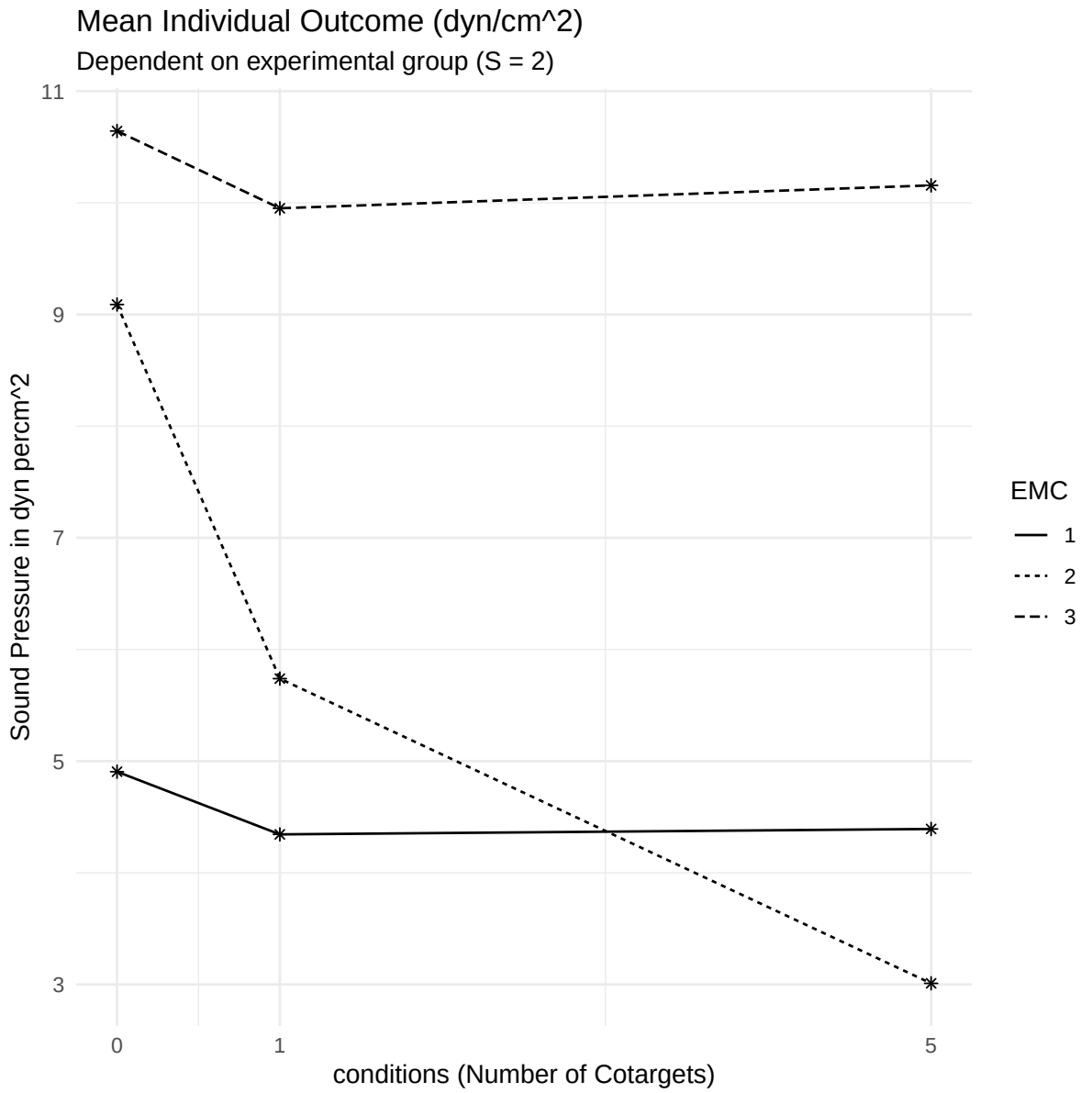
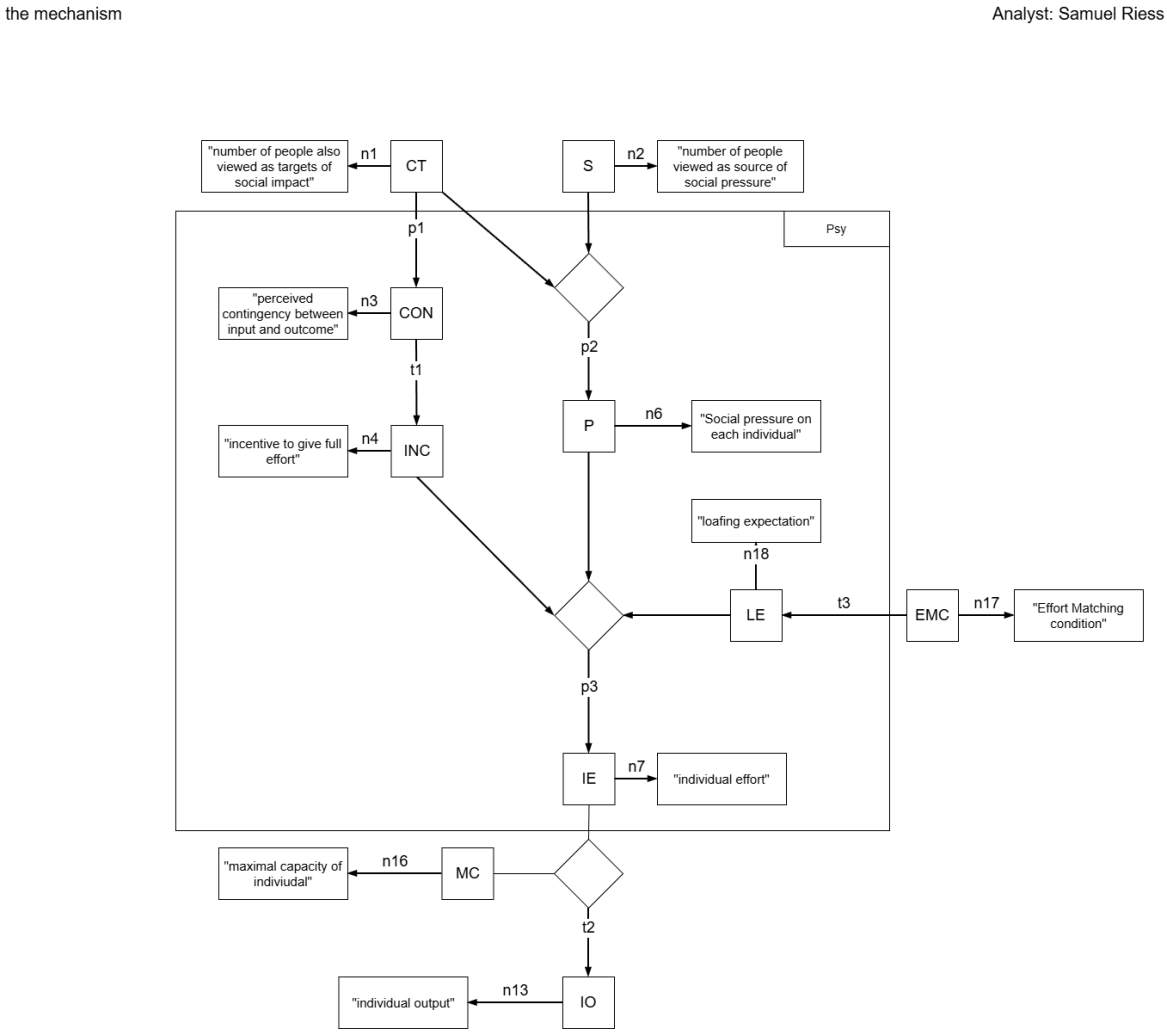


Figure 12

Structure of the loafing mechanism without Fallacies